



FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	EggTech-00000000253
Award Program	Egg Tech Prize Phase I
Project Title	Orbem.ai: magnetic resonance imaging meets artificial intelligence for automated and non-invasive in ovo sex determination
Grant Start Date	09/01/2019
Grant End Date	08/31/2021
Total Grant Budget	\$1,149,911.11
Total FFAR Budget	\$400,000.00
Matching Funder(s)	Funds matched internally by company
Final Report Date	08/28/2023
Project Director/Principal Investigator Information	
Full Name	Dr. Pedro Gomez
Email	pedro.gomez@orbem.ai
Phone	+49 89 954 597 540
Grantee Organization Information	
Organization Name	Orbem GmbH
Address	Wilhelmine-Reichard-Straße 7
City, State Zip	Munich, D-80935 Germany



Tax ID	DE327553341
Authorized Signing Official (ASO) Information	
Name	Dr. Pedro Gomez
ASO Title	CEO
Email	pedro.gomez@orbem.ai
Phone	+49 89 954 597 540

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Munich, Germany

- 1.2. How many new jobs were created because of this grant funding?

4

- 1.3. How many jobs were maintained because of this grant funding?

4

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.



Orbem combines accelerated magnetic resonance imaging (MRI) with advanced artificial intelligence (AI) to determine the sex of developing embryos in-ovo. Accelerated MRI enables us to measure structures and physiological processes within eggs in a non-invasive manner, whereas AI processes the data to automatically predict the sex of developing embryos. With this novel technology, our objective was to perform accurate, high-speed, and early-stage sex determination in a contactless and safe procedure.

Our non-invasive technology offers numerous advantages over existing approaches. First, it functions prior to chicks developing sensation. Second, it operates with high throughput, as our setup enables the parallel measurement of multiple eggs. Third, it is effective across all breeds, does not harm or modify the eggs in any way, and has no impact on hatchability or chick welfare.

The development of this technology is relevant to address needs of the U.S. food and agriculture as it can revolutionize egg production by preventing the culling of day-old male chicks in poultry hatcheries and improve the efficiency and sustainability of egg production.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The following key findings were made throughout this project:

1. It is technically feasible to determine the sex of an embryo in-ovo using a combination of accelerated MRI and advanced AI techniques.
2. The day of sex determination ranges from days 12-15.
3. As of the end of the grant period (August 2021) in-ovo sex determination on days 12-15 sufficed to meet the requirements of upcoming regulations in France.
4. As of the the writing of this report (August 2023), it has furthermore been scientifically established that embryos **do not perceive pain** prior to day 13¹, rendering an in-ovo sexing technology operating on day 12 fully aligned with animal welfare standards.
5. The in-ovo sexing technology based on AI and MRI is completely non-invasive, works for all breeds, and is scalable such that it can be implemented by U.S. hatcheries in the near future.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to

¹ <https://www.bmel.de/SharedDocs/Pressemitteilungen/DE/2023/051-verbot-kuekentoeten.html>



answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The main research outcome is the demonstration of the technical feasibility of in-ovo sexing technology. This technology is non-invasive, functions for all breeds, and prior to the onset of pain perception. With further development, the technology can be brought to technical maturity and deployed at large-scale at every layer hatchery in the U.S. With a modular setup, the system will be capable of processing up to 24,000 eggs per hour with high accuracy.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Our two-year project followed three specific aims:

- 1. Hardware development to build a prototype for high-throughput, non invasive egg scanning (completed by October 30th, 2020)
- 2. MRI acceleration software development to increase current throughput from 900 eggs per hour to at least 2,000 eggs per hour (completed by August 31, 2021)
- 3. AI algorithm development to improve classification accuracy from current 78% to a minimum of 98% accuracy (not completed within the timeframe of the grant, delayed and completed by January 1st, 2023)

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

- 2.6. What was accomplished under the goals/specific aims or milestones for the



project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

We completed each of the specific aims as follows.

Objective 1: Build hardware prototype

We successfully accomplished this objective by performing the following key activities:

1. Definition of all technical requirements for imaging, including magnetic field strength, gradient performance, and imaging parameters
2. Definition of all technical requirements for automation, including conveyor belt compatibility with MRI, effective line speed and acceleration.
3. Sourcing of key strategic partners to deliver MRI equipment and automation equipment.
4. Design and implementation of coil elements such that they are as close as possible to the imaged object
5. Installation of prototype system, including system and software integration of MRI instrument with automation equipment.

A key finding in objective 1 was the establishment of the first-ever MRI-compatible egg conveyor belt and the determination of all the main technical parameters to enable high-throughput egg scanning. The development of this prototype was instrumental for us to validate the technical requirements for next-generation product development.

Objective 2: Demonstrate 2,000 egg per hour throughput

We successfully accomplished this objective by performing the following key activities:

1. Definition, development, and implementation of core MRI pulse sequence, including establishment of imaging parameters such as field of view, bandwidth, excitation pulse characteristics, echo time, repetition time, and flip angle.
2. Technical implementation of reconstruction techniques on a high-performance computational unit on premise.
3. Development and deployment of software data flows that enable quasi real-time image acquisition, reconstruction, and inference.
4. Software integration of automation equipment to enable the entire workflow,



including scanning, moving and sorting.

5. Successful demonstration of 2,000 eggs scanned within one hour, with each image successfully tracked and uploaded into our database.

Achieving objective 2 is the most important accomplishment of this grant, especially as it also required a successful prototype integration (objective 1). By achieving this objective, we successfully demonstrated the technical feasibility of high-throughput MRI scanning without any human intervention. This laid the groundwork to fully automate the classification process in pursuit of objective 3.

Objective 3: Achieve 98% classification accuracy

We *did not* successfully achieve this objective within the timeframe of the grant. With the Coronavirus outbreak a few months into the grant, supply chains were completely disrupted, delaying the completion of objective 1 (prototype installation) and objective 2 (accelerated imaging). As objective 3 necessitated both objectives to be completed to fully carry out all key activities, we were not able to complete this objective within the 2-year timeframe established by the grant agreement. Nonetheless, we focused on the following key activities in preparation for the completion of this objective:

1. Establishment of the data infrastructure to be able to efficiently train models in the cloud.
2. Development and implementation of state-of-the art convolutional and recurrent neural network architectures for classification of 3D data tensors.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

The most important aspect regarding the scientific rigor of our approach is to validate the results of our classification algorithms. That is, we needed to establish a method to determine if the eggs we classified as male or female are *actually* male or female. To do this, we established a process to collect ground truth data from independent measurements. The independent measurements, either blood sampling followed by PCR tests or morphological examination of the embryos, ensured we obtained robust and unbiased results. These reference measurements were also used in the process of developing and training our AI models.

- 2.8. List key stakeholders that could be served by results of this project.

Layer Hatcheries - layer hatcheries benefit from a solution to stop the culling of one day old males



Legislators - Legislators are key to enacting laws that favor animal welfare and benefit from the results of this project

NGOs - NGOs can use the results of this project to influence legislation and increase animal welfare

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The main challenge we faced was the onset of the Coronavirus epidemic (see Section 2.6). This caused a significant delay in our development timelines due to disrupted supply chains, resulting in us not achieving the expected accuracy throughout the grant period. Nonetheless, using external funding we further developed our solution outside the grant period and obtained the desired results.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)



Nothing to report

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Nothing to report

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Nothing to report

- 3.2. Please provide a list of citations for the information products produced during the project.

Nothing to report

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Nothing to report

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Nothing to report

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No



- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

N/A

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

No

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

N/A

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

N/A



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Pedro Gomez

PI Signature:  Date: 30-oct-2023

Authorized Signing Official (ASO) Name: Pedro Gomez

ASO Signature:  Date: 30-oct-2023